

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

BASIC MATHEMATICS

DIBY (DO IT BY YOURSELF)

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DIBY-01



The multiplication of a rational number ' x ' and an irrational number ' y ' is

- A** Always rational
- B** Rational except when $y = \pi$
- C** Always irrational
- D** Irrational except when $x = 0$

Ans: (D)

DIBY-02

If a, b, c are real, then $a(a - b) + b(b - c) + c(c - a) = 0$, only if

- A** $a + b + c = 0$
- B** $a = b = c$
- C** $a = b$ or $b = c$ or $c = a$
- D** $a - b - c = 0$

Ans: (B)

DIBY-03

If a, b, c are real and distinct numbers, then the value of $\frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a-b)(b-c)(c-a)}$

- A** 1
- B** abc
- C** 2
- D** 3

Ans: (D)

DIBY-04

If $3a - 2b + c = 0$, then find the value of $9a^2 - 4b^2 + c^2 + 6ac$ is

- ☐ A -1
 ☐ B 1
 ☐ C 2
 ☐ D 0

Ans: (D)

DIBY-05

If x, y, z are all different real numbers, then prove that

$$\frac{1}{(x-y)^2} + \frac{1}{(y-z)^2} + \frac{1}{(z-x)^2} = \left(\frac{1}{x-y} + \frac{1}{y-z} + \frac{1}{z-x} \right)^2$$

DIBY-06

If $x = 2^{\frac{2}{3}} + 2^{\frac{1}{3}}$ then value of $x^3 - 6x =$ _____

Ans: (6)

DIBY-07

If $x^2 - 4x + 1 = 0$, then what is the value of $x^3 + \frac{1}{x^3}$?

Ans: (52)

DIBY-08

If $x = 1 + 2^{\frac{1}{3}} + 2^{\frac{2}{3}}$, then $x^3 - 3x^2 - 3x = \underline{\hspace{2cm}}$

Ans: (1)

DIBY-09

If $x + \frac{1}{x} = a$, then what is the value of $x^3 + x^2 + \frac{1}{x^3} + \frac{1}{x^2}$?

Ans: $a^3 + a^2 - 3a - 2$

Column-I

(A) $\frac{5x-3}{-4} \leq 8$

(B) $\frac{x-1}{x+2} > 0$

(C) $\frac{(x+3)(x-2)}{(x-5)} \leq 0$

(D) $\frac{(x+4)(x+8)}{(x-9)(x+1)(x-3)} > 0$

Column-II

(P) $x \in (-\infty, -3] \cup [2, 5)$

(Q) $x \in (-8, -4) \cup (-1, 3) \cup (9, \infty)$

(R) $x \in \left[-\frac{29}{5}, \infty\right)$

(S) $x \in (-\infty, -2) \cup (1, \infty)$

Ans: A $\rightarrow$ R, B $\rightarrow$ S, C $\rightarrow$ P, D $\rightarrow$ Q

DIBY-11



Find sum of all integral values of x satisfying $\frac{x^2 - 5x + 6}{x^2 - x - 12} \leq 0$

[Ans: 03]

DIBY-12

Solve: $\frac{x}{x-2} > \frac{1}{x+3}$

[Ans. : $x \in (-\infty, -3) \cup (2, \infty)$]

DIBY-13

Solve: $\frac{2x+3}{x^2+x-12} \leq \frac{1}{2}$

[Ans. : $x \in (-\infty, -4) \cup [-3, 3) \cup [6, \infty)$]

DIBY-14

Solve : $(x + 5)(2x - 3)^5(-x + 7)^3(3x + 8)^2 < 0$

$$\left[\text{Ans. : } x \in \left(-5, -\frac{8}{3}\right) \cup \left(-\frac{8}{3}, \frac{3}{2}\right) \cup (7, \infty) \right]$$

DIBY-15

Solve : $(x^2 - 3x + 2)(x^3 - 3x^2)(4 - x^2) \geq 0$

$$[\text{Ans. : } x \in (-\infty, -2) \cup \{0\} \cup [1, 3]]$$

DIBY-16

Solve : $\frac{1}{x^2 + x} \leq \frac{1}{2x^2 + 2x + 3}$

$$[\text{Ans. : } x \in (-1, 0)]$$

$$(1) \quad \frac{6x-5}{4x+1} < 0$$

$$[\text{Ans. } (-1/4, 5/6)]$$

$$(2) \quad \frac{3}{x-2} < 1$$

$$[\text{Ans. } (-\infty, 2) \cup (5, \infty)]$$

$$(3) \quad \frac{1}{x-1} \leq 2$$

$$[\text{Ans. } (-\infty, 1) \cup [3/2, \infty)]$$

$$(4) \quad \frac{x-1}{x} - \frac{x+1}{x-1} < 2$$

$$[\text{Ans. } (-\infty, -1) \left(0, \frac{1}{2}\right) \cup (1, \infty)]$$

$$(5) \quad 1 + \frac{2}{x-1} \geq \frac{6}{x}$$

$$[\text{Ans. } (-\infty, 0) \cup (1, 2] \cup [3, \infty)]$$

DIBY-18



Solve:
$$\frac{(x-1)^2(x^2-4)(x^2-5x+6)}{(x^3+6x^2)} \geq 0$$

[Ans. $x \in (-6, -2] \cup \{1, 2\} \cup [3, \infty)$]

DIBY-19

The greatest negative integer satisfying $x^2 - 4x - 77 < 0$ and $x^2 > 4$, is

[Ans: -3]

DIBY-20

The solution set of $x^2 + 2 \leq 3x \leq 2x^2 - 5$, is

(A) No Solution

(B) $[1, 2]$

(C) $(-\infty, -1] \cup [5/2, \infty)$

(D) none of these.

[Ans: A]

DIBY-21

Solve: $-1 < \frac{3x-1}{x-2} \leq 4$

Ans: $x \in \left(-\infty, \frac{3}{4}\right) \cup [7, \infty)$

DIBY-22

Values of x simultaneously satisfying: $x^2 - x - 6 \geq 0$ & $x^2 - 4x < 0$

Ans: $x \in [3, 4]$

DIBY-23

Solve: $(x^2 - x - 1)(x^2 - x - 7) < -5$

Ans: $x \in (-2, -1) \cup (2, 3)$

DIBY-24



Solution of $\frac{(x^4)(x+1)^2(x-2)}{(x-3)^3(x+4)} > 0$ is

(A) $x \in (-4, 2) \cup (4, \infty)$

(B) $x \in (-4, -1) \cup (-1, 2) \cup (3, \infty)$

(C) $x \in (-4, 0) \cup (0, 2) \cup (3, \infty)$

(D) $x \in (-4, -1) \cup (-1, 0) \cup (0, 2) \cup (3, \infty)$

[Ans: D]

DIBY-25 (Challenger Question)

Solve: $\frac{(x^3 - 4x^2 + 3x - 12)^{18}(5 - 7x)^{19}}{(4 - x^2)^{20}(x + 2)^{21}(x + 3)^{22}} \leq 0$

[Ans: $x \in (-\infty, -2) \cup \left[\frac{5}{7}, \infty\right) - \{-3, 2\}$]

DIBY-26

Solve following Inequalities :

$$(i) \quad \left(\frac{x^4 + x^2 + 1}{x^2 - 11x + 28} \right) \geq 0$$

$$\text{Ans. } x \in (-\infty, 4) \cup (7, \infty)$$

$$(ii) \quad \frac{x^2 - x + 1}{x^2 + 7x + 15} \geq 0$$

$$\text{Ans. } x \in R$$

DIBY-27

The number of real solution of the equation, $x^2 - |x| - 12 = 0$ is

[JEE Mains-2021]

(A) 2

(B) 3

(C) 1

(D) 4

DIBY-28

Solve $|x^2 + 4x + 3| + 2x + 5 = 0$

[JEE-1987, 5M]

Ans. : $x = -4$ and $(-1 - \sqrt{3})$

DIBY-29

The sum of all the real roots of the equation $|x - 2|^2 + |x - 2| - 2 = 0$ is _____.

[JEE-1997, 2M]

Ans. : 4

DIBY-30

$(x - 3)^2 + |x - 3| - 11 = 0$, the sum of solutions of this equation is?

- (A) 5 (B) 6 (C) 7 (D) 8

[JEE Mains-2021]

Ans. (B)

DIBY-31

The sum of all roots of the equation $|x^2 - 8x + 15| - 2x + 7 = 0$ is

- (A) $9 - \sqrt{3}$ (B) $9 + \sqrt{3}$ (C) $11 - \sqrt{3}$ (D) $11 + \sqrt{3}$ Ans. (B)

[JEE Mains-2023]

DIBY-32

Number of real solutions of $|x - 1| = |x - 2| + |x - 3|$ is

- (A) 0 (B) 1 (C) 2 (D) more than 2

Ans. (C)

DIBY-33

Solve for x :

(i) $|||x + 5| - 3| - 1| = 2$

(ii) $|||x - 5| - 7| - 3| - 2| = 1$

Ans. (i) $x \in \{-11, -5, -1\}$

(ii) $x \in \{-8, -6, -2, -4, 0, 2, 4, 6, 8, 10, 12, 14, 16, 18\}$

DIBY-34

Solution of Modulus equation: $|x - 3| + 2|x + 1| = 4$

Ans. ($x = -1$)

DIBY-35



Solve: $\left| \frac{x-3}{x+1} \right| \leq 1$

Ans. $x \geq 1$

DIBY-36

Solve: $\left| 1 + \frac{3}{x} \right| > 2$

Ans. $x \in (-1, 0) \cup (0, 3)$

DIBY-37

Number of non-positive integral values of ' x ' satisfying the given inequality,

$|x^2 - 1| \leq |2x - 1|$ is

(A) 0

(B) 1

(C) 2

(D) 3

Ans. (D)

DIBY-38

Solve for x :

$$(i) \quad \frac{(|x|-1)}{(|x|-2)} \leq 0$$

$$(ii) \quad \frac{(|x|-1)(|x|-3)}{|x|^2-2|x|} \geq 0$$

$$(iii) \quad \frac{(|x|^2-5|x|+6)}{(4-|x|^2)} \leq 0$$

Ans.

$$(i) \quad x \in (-2, -1] \cup [1, 2)$$

$$(ii) \quad x \in (-\infty, -3] \cup (-2, -1] \cup [3, \infty) \cup [1, 2)$$

$$(iii) \quad x \in (-\infty, -3] \cup [3, \infty)$$

DIBY-39

Solve for x :

$$(i) \quad \frac{1}{2} \leq |2x-1| \leq \frac{3}{5}$$

$$(ii) \quad -\frac{1}{3} \leq |3x-4| \leq 2$$

$$(iii) \quad 2 \leq |4-5x| \leq \frac{10}{3}$$

$$(iv) \quad 3 < |x^2-1| < 8$$

Ans.:

$$(i) \quad x \in \left[\frac{1}{5}, \frac{1}{4}\right] \cup \left[\frac{3}{4}, \frac{4}{5}\right]$$

$$(ii) \quad x \in \left[\frac{2}{3}, 2\right]$$

$$(iii) \quad x \in \left[\frac{2}{15}, \frac{2}{5}\right] \cup \left[\frac{6}{5}, \frac{22}{15}\right]$$

$$(iv) \quad x \in (-3, -2) \cup (2, 3)$$

DIBY-40

Solve if $|x - 5| + |x + 4| = 9$

- (A) $[-4, 5]$ (B) $(-4, 5)$ (C) $(-4, 5]$ (D) $[-4, 5)$

Ans. (A)

DIBY-41

Solve: $\frac{|x-2|-3}{|x+5|+2} \geq 0$.

Ans.: $x \in (-\infty, -1] \cup [5, \infty)$

DIBY-42

Solve: $\left| \frac{3}{x-5} \right| > 1$.

Ans.: $x \in (2, 8) - \{5\}$

DIBY-43

Solve: $|x^2 - 3x - 3| > |x^2 + 7x - 13|$.

Ans.: $x \in (-\infty, -4) \cup (1, 2)$

DIBY-44



Number of solution of the equation $2\sqrt{x^2 + 2x + 1} - 3\sqrt{x^2 - 4x + 4} + |5 - x| = 5$

- (A) 0 (B) 1 (C) 2 (D) More than 2

Ans: C

DIBY-45

If $m_1 = \log_8 16$, $m_2 = \log_{81} 27$, $m_3 = \log_{1/3} 1/9$, $m_4 = \log_{1/3} 9\sqrt{3}$, then $m_1 \cdot m_2 \cdot m_3 \cdot m_4$

Ans: -5

DIBY-46

The value of $81^{(1/\log_5 3)} + 27^{\log_9 36} + 3^{4/\log_7 9}$ is equal to

- (A) 49 (B) 625 (C) 216 (D) 890

Ans: D

DIBY-47

Let $N = \frac{\log_3 135}{\log_{15} 3} - \frac{\log_3 5}{\log_{405} 3}$. Then N is

- (A) a natural number (B) a prime number
(C) a rational number (D) an integer

Ans: A,B,C,D

DIBY-48

Greatest integer less than or equal to the number $\log_2 15 \cdot \log_{1/6} 2 \cdot \log_3 1/6$ is

- (A) 4 (B) 3 (C) 2 (D) 1

Ans: C

DIBY-49

$\frac{1}{1+\log_b a + \log_b c} + \frac{1}{1+\log_c a + \log_c b} + \frac{1}{1+\log_a b + \log_a c}$ has the value equal to

- (A) abc (B) $\frac{1}{abc}$ (C) 0 (D) 1

Ans: D

DIBY-50

$\frac{1}{\log_{\sqrt{bc}} abc} + \frac{1}{\log_{\sqrt{ca}} abc} + \frac{1}{\log_{\sqrt{ab}} abc}$ has the value equal to

- (A) $1/2$ (B) 1 (C) 2 (D) 4

Ans: B

DIBY-51

If $\log_7 2 = m$, then $\log_{49} 28$ is equal to

- (A) $2(1 + 2m)$ (B) $\frac{1+2m}{2}$ (C) $\frac{2}{1+2m}$ (D) $1 + m$

Ans: B

DIBY-52

Value of $\log_2 \left(\log_2 \left(\log_3 (\log_3 27^3) \right) \right) =$

Ans: 0

DIBY-53

Value of x satisfying $x^2 + 7^{\log_7 x} - 2 = 0$

Ans: ($x = 1$)

DIBY-54

Match the column

Column-I

- (A) If $\log a = 2\log b$, then
- (B) If $\log a + 2\log b = 0$, then
- (C) If $3\log a - 2\log b = 1$, then
- (D) If $\ln\left(\frac{a+b}{2}\right) = \frac{1}{2}(\ln a + \ln b)$, then

Column-II

- (P) $a^3 = 10b^2$
- (Q) $a = b$
- (R) $a = b^2$
- (S) $ab^2 = 1$

Ans: A-R, B-S, C-P, D-Q

DIBY-55

The sum of all the solutions of the equation $2\log x - \log(2x - 75) = 2$

- (A) 30
- (B) 350
- (C) 75
- (D) 200

Ans: D

DIBY-56

The solution of the equation $\log_7 \log_5 \left(\sqrt{x^2 + 5} + x \right) = 0$.

- (A) $x = 2$ (B) $x = 3$ (C) $x = 4$ (D) $x = -2$

Ans: C

DIBY-57

Number of real solution of $\log_5 [2 + \log_3 (x + 3)] = 0$ is

Ans: 1

DIBY-58

$$\log_7 \left(\frac{2x-6}{2x-1} \right) \geq 0$$

Ans: $\left(-\infty, \frac{1}{2} \right)$

DIBY-59

If $\log_{0.5} \log_5(x^2 - 4) > \log_{0.5} 1$, then ' x ' lies in the interval

- | | |
|--|--|
| (A) $(-3, -\sqrt{5}) \cup (\sqrt{5}, 3)$ | (B) $(-3, -\sqrt{5}) \cup (\sqrt{5}, 3\sqrt{5})$ |
| (C) $(\sqrt{5}, 3\sqrt{5})$ | (D) ϕ |

Ans: A

DIBY-60

Solution set of the inequality $2 - \log_2(x^2 + 3x) \geq 0$ is

- | | |
|--------------------------------------|--------------------------------------|
| (A) $[-4, 1]$ | (B) $[-4, -3] \cup (0, 1]$ |
| (C) $(-\infty, -3) \cup (1, \infty)$ | (D) $(-\infty, -4) \cup [1, \infty)$ |

Ans: B

DIBY-61

Find the sum of all integral values of x satisfying $(\log_5 x)^2 + \log_{5x} \left(\frac{5}{x} \right) = 1$

Ans. 6

DIBY-62

If $A = \frac{(\log_3 1 - \log_3 4)(\log_3 9 - \log_3 2)}{(\log_3 1 - \log_3 9)(\log_3 8 - \log_3 4)}$, then find the value of $2(3^A)$

Ans. 9

DIBY-63

values of x for which $y = \log_{x-10}(x^2 - 6x + 8)$ is defined is

Ans. $x \in (10, \infty) - \{11\}$

DIBY-64

If $\left(\frac{a}{b}\right)^{x-1} = \left(\frac{b}{a}\right)^{x-3}$, then the value of x is

(A) $\frac{1}{2}$

(B) 1

(C) 2

(D) $\frac{7}{2}$

Ans: C

DIBY-65

$\sqrt{\left(\frac{5}{7}\right)^{6x-4}} = \frac{2401}{875}$, find value of x

Ans: -1/3

DIBY-66

The set of all solutions of the inequality $\left(\frac{1}{2}\right)^{x^2-2x} < \frac{1}{4}$ contains the set

(A) $(-\infty, 0)$

(B) $(-\infty, 1)$

(C) $(1, \infty)$

(D) $(3, \infty)$

Ans: D

DIBY-67

The solution set of inequality $\log_{(3x^2+1)} 2 < \frac{1}{2}$

- (A) $|x| > 1$ (B) $|x| < 1$ (C) ϕ (D) None of these

Ans: A

DIBY-68

If $N = 7^{\log_4 900}$, $A = 2^{\log_2 4} + 3^{\log_2 4} + 4^{\log_2 2} - 4^{\log_2 3}$

$D = (\log_5 49)(\log_7 125)$

Find $P = \log_{\left(A - \frac{N}{10}\right)} |N + A + D + 6| - \log_5 2$.

Ans: 0

DIBY-69

Let $3^a = 4$, $4^b = 5$, $5^c = 6$, $6^d = 7$, $7^e = 8$ and $8^f = 9$. The value of the product $(abcdef)$ is:

- (A) 1 (B) 2 (C) $\sqrt{6}$ (D) 3

Ans: B

DIBY-70



If $A = \log_{\sqrt{5}} \left(\left(5^{\frac{1}{2}} \right)^{\frac{1}{2}} \right)^{\frac{1}{2}} \cdot 10$ times, then value of $\log_{\sqrt{3}}(1024A + 1)$, is equal to

(A) 1 (B) 3 (C) 5 (D) 2

Ans: D

DIBY-71

The number of positive integers not satisfying the inequality

$$\log_2(4^x - 2 \cdot 2^x + 17) > 5.$$

(A) 2 (B) 3 (C) 4 (D) 1

Ans: A

DIBY-72: Acidic

$$\text{Solve : } 2\log_2\log_2x + \log_{1/2}\log_2(2\sqrt{2}x) = 1$$

Ans. 8

DIBY-73: Challenger

$$\text{Solve : } \log_{x^2+6x+8}(\log_{2x^2+2x+3}(x^2 - 2x)) = 0$$

Ans. -1

Column-I		Column-II	
A.	$5^x - 24 = \frac{25}{5^x}$	p.	-3
B.	$(2^{x+1})(5^x) = 200$	q.	-2
C.	$4^{2/x} - 5(4^{1/x}) + 4 = 0$	r.	-1
D.	$2^{2x+1} - 33(2^{x-1}) + 4 = 0$	s.	0
E.	$\frac{2^{x-1} \cdot 4^{x+1}}{8^{x-1}} = 16$	t.	1
F.	$3^{2x+1} + 10(3^x) + 3 = 0$	u.	2
G.	$64(9^x) - 84(12^x) + 27(16^x) = 0$	v.	3
H.	$5^{2x} - 7^x - 5^{2x}(35) + 7^x(35) = 0$	w.	None

Ans:**A → u;****B → u;****C → t;****D → q, v;****E → p, q, r, s, t, u, v;****F → w;****G → t, u;****H → s**

*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**